



YOUR TECHNOLOGY & INNOVATION PARTNER

Advanced AI Training
Inventory Management with AI-Powered
Demand Forecasting

Course Objectives

- Gain a comprehensive understanding of the current AI landscape in forecasting in reducing stockouts and overstock.
- Collect, clean, and transform historical sales and inventory data, including external influencing factors (e.g., promotions, holidays, trends).
- Explore the provided workshop dataset, understanding its structure, relevance, and predict demand for raw materials and finished goods.
- Clarify the requirements and expectations for the group project, Incorporate variables like marketing campaigns, weather, and seasonality to enhance forecasting performance.
- Develop skills to design and implement AI-driven forecasting systems tailored to customer preferences and behaviors.
- Calculate and interpret metrics such as RMSE (Root Mean Square Error) and MAPE (Mean Absolute Percentage Error) to assess model accuracy.

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AI in Inventory Management

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Automated Replenishment

3

Real-Time Monitoring

4

Adaptability



Group Project Guideline



Overview

- This project aims to provide learners with hands-on experience in building an AI-powered demand forecasting system for inventory management in a manufacturing or retail context.
- By utilizing publicly available datasets, learners will explore the complete AI development lifecycle—from data collection, preprocessing, and feature engineering to designing, training, and evaluating time-series forecasting models such as Prophet or LSTM.
- The project will include performance evaluation using industry-standard metrics like RMSE (Root Mean Square Error) and MAPE (Mean Absolute Percentage Error), and culminate in the deployment of a real-time inventory monitoring dashboard with demand forecasts and restocking alerts.
- The ultimate goal is to build a practical forecasting solution that optimizes stock levels, minimizes overstock and stockouts, and supports smarter procurement and production planning through data-driven insights.

Datasets

Learners may either **use a dataset of their choice** or select from one of the recommended datasets listed below:

Favorita Grocery Sales Forecasting

Includes product families sold data (store_nbr, sales) along with product and category information.

M5 Forecasting

Walmart sales data at the SKU level

Rossmann Store Sales

Sales and promotion data from Rossmann drug stores

UCI Online Retail Dataset

Transactions from an online UK-based retailer.

Dataset

Favorita Grocery Sales Forecasting Dataset

<https://www.kaggle.com/competitions/store-sales-time-series-forecasting/data>

Contains daily sales data from a grocery store chain in Ecuador, with external factors like promotions and holidays.

Table data (3 types of events, namely “store_nbr”, “sales” and “onpromotion”)

Descriptive

Diagnostic

Predictive

Prescriptive

Predict next-day or next-week sales per item per store → use this for auto-restocking alerts.



Dataset

M5 Forecasting – Accuracy Dataset dataset

<https://www.kaggle.com/competitions/m5-forecasting-accuracy/data>

Walmart sales data at the SKU level, with calendar and price data across stores and categories.

Fields: `item_id`, `dept_id`, `store_id`, `sales`, `calendar`, `sell_price`

Weekly/daily demand forecasting with external regressors

Sentiment Analysis

Forecast weekly demand across categories → plan inventory purchases across stores.



Dataset

Rossmann Store Sales dataset

<https://www.kaggle.com/competitions/rossmann-store-sales/data>

Commerce Behavior Dataset offers a broader view of user behavior—including cart removals and session flow—making it ideal for full journey analysis and funnel tracking.

Table data (4 types of Sales, Promo, “cart”, Customers and CompetitionDistance (e.g., category, Open, StateHoliday).)

Descriptive

Diagnostic

Predictive

Prescriptive

Estimate demand per store

Optimize warehouse stock allocation and delivery



Dataset

UCI Online Retail dataset

<https://archive.ics.uci.edu/dataset/352/online+retail>

This is a transactional data set which contains all the transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based and registered non-store online retail.

Invoice timestamps, quantity, country (external calendar/holiday data can be added)

**Descriptiv
e**

Diagnostic

Predictive

Prescriptive

Forecast weekly sales → automate replenishment and reduce deadstock.



Project Guideline

Business Understanding

1

Define Problem and Pain Points

Defines a Real Business Problem

Demonstrates User Empathy

2

Define Solutions and Methods

Presents a Clear, Practical Solution

Justifies the Approach

3

Define Success Criteria

Defines Clear Success Metrics

Aligns with Stakeholder Impact

Data Understanding

1

Describe data chosen

Describes Data Thoroughly

Justifies Data Relevance

2

Explore the data chosen

Applies Diverse Data Exploration Techniques

Demonstrates Data Understanding

3

Identify key findings and patterns

Discovers Meaningful Insights

Links Insights to Business Goals

4

Demo codes for Data Exploration

Delivers Clean and Reusable Code

Ensures Reproducibility and Clarity

Data Preparation

1

Describe and explain

Executes and Justifies Data Cleaning

Transforms Data for Business & Modeling Use

2

Demo codes for Data Preparation

Implements Clean, Reusable Code

Ensures Robustness and Reproducibility

Modelling

1

Describe and explain

Justifies Model Choice Clearly

Understand pain points & Identify goals

2

Analyze pros and cons

Evaluates Key Trade-offs

Balances Business & Technical Needs

3

Describe and explain

Explains Full Modeling Workflow

Demonstrates Mastery of Model Lifecycle

4

Demo codes

Delivers Modular & Reproducible Code

Ensures Readability & Robustness

Evaluation

1

Evaluate results

Evaluates Model Performance Thoroughly

Links Results to Business Goals

2

Review process

Reviews Workflow Transparently

Identifies Areas for Improvement

3

Determine next steps

Outlines Practical Next Steps

Shows Strategic Foresight

4

Demo codes

Produces Clear, Modular Code with Evaluation

Applies Robust Validation Techniques

Deployment

1

Plan deployment

Outlines Complete Deployment Strategy

Ensures Scalability and Business Alignment

2

Plan monitoring and maintenance

Establishes Robust Monitoring Plan

Prepares for Maintenance and Recovery

3

Review project

Conducts Thoughtful Project Reflection

Identifies Lessons Learned for Future Use

Other

1

Reference and Citation

Uses Consistent Citation Style

Includes Full References

2

Report with clear format

Presents Information in a Clear, Structured Format

Enhances Readability with Visuals and Styling

Conclusion

- This project demonstrated the practical application of AI and machine learning techniques to address real-world challenges in inventory management. By leveraging time-series forecasting models such as **Prophet and LSTM**, we showed how historical sales data and external factors—like promotions and holidays—can be used to accurately predict product demand.
- Through experimentation with multiple publicly available datasets, we explored various forecasting scenarios, including retail sales, store-level predictions, and behavioral demand trends. The models were evaluated using standard performance metrics like **RMSE and MAPE**, offering insights into their reliability and accuracy in different business contexts.
- The final output showcases how AI can transform inventory planning by delivering demand trends, restocking alerts, and actionable insights in real time. This not only helps reduce stockouts and overstock but also supports data-driven decision-making for supply chain and operations teams.
- Overall, this hands-on project has equipped learners with end-to-end experience in building and deploying AI-powered inventory forecasting systems, paving the way for smarter, more efficient inventory management solutions in any industry.

Thank you